National higher school of advanced technologies

Academic year: 2026–2027 | first-year master's degree

Machine learning — practical session 2

Python for standardized PCA

Objective: Implement Principal Component Analysis (PCA) from scratch using Python and compare the results with the predefined PCA implementation provided by Scikit-learn.

Exercise :

Consider the following dataset containing the grades of six students in four subjects.

Student	Mathematics	Statistics	Programming	Algorithms
S1	12	11	14	13
S2	15	14	16	15
S3	8	9	10	9
S4	17	16	18	17
S5	10	12	11	10
S6	14	15	13	14

Part A: Standardized PCA implementation from scratch

1. Load the dataset into a Pandas DataFrame.
2. Display the dimensions and descriptive statistics of the dataset.
3. Standardize the numerical variables using the mean and standard deviation.
4. Compute the correlation matrix of the standardized data.
5. Calculate the eigenvalues and eigenvectors of the covariance matrix using NumPy.
6. Sort the eigenvalues in descending order and reorder the corresponding eigenvectors.
7. Select the two eigenvectors associated with the largest eigenvalues.
8. Project the standardized data onto the two selected principal components.
9. Calculate the explained variance ratio of each component and the cumulative explained variance.

Part B: Standardized PCA using Scikit-learn

10. Apply standardized PCA using the predefined standardized PCA class from sklearn.decomposition.
11. Reduce the dataset to two principal components.
12. Display the transformed coordinates of the students.
13. Calculate the explained variance ratio and cumulative explained variance.

Part C: Comparison and interpretation

14. Compare the principal component coordinates obtained from both implementations.
15. Compare the eigenvalues and explained variance ratios.
16. Plot the students in two dimensions using both implementations.
17. Explain any differences observed between the results, particularly possible sign changes in the principal components.
18. Determine the percentage of total variance retained by the first two principal components.

Deliverables

Submit:

- The Python implementation of standardized PCA from scratch.
- The implementation using Scikit-learn.
- A comparison of the numerical results.
- The visualizations and a brief interpretation of the findings.